

Multimodal Search Alliance

A Multi-Stage Retrieval System for Trading Strategy Discovery

DEEP LEARNING FOR SEARCH – COURSE PROJECT

BM25

SPLADE

BI-ENCODER

CROSS-ENCODER

FAISS

LAMBDAMART

INFONCE

AGENTIC CRAG

Value Proposition

What is the value proposition? Who might need this?

⚠️ PROBLEM

Quantitative traders and portfolio managers need to discover relevant trading strategies from a corpus of **205,000+ documents**, but single-retriever systems suffer from the **lexical gap**, missing **multimodal signals**, and lack **user feedback integration**.

💡 SOLUTION

A **4-phase multimodal retrieval alliance** that fuses sparse (BM25, SPLADE) and dense (text Bi-encoder, curve CNN) retrieval via **Reciprocal Rank Fusion**, reranks with **LambdaMART**, and applies **Agentic CRAG** for user constraint refinement.

205K+

documents

4

retrieval systems

<200ms

latency

TARGET USERS

Hedge funds · Quant desks · Algo trading platforms · Academic researchers

System Architecture: The Alliance Pipeline

PIPELINE OVERVIEW

1 RECALL

BM25 Inverted Index (L02)
SPLADE Learned Sparse (L05)
Dense Text Bi-encoder (L06)
Dense Curve 1D-CNN

RRF Fusion (k=60)
→ Top-100 candidates



2 FEATURES

6 LTR features per candidate:

bm25_score
splade_cos
dense_text_cos
dense_curve_cos
diff_sharpe
diff_drawdown

Feature extraction (L09)



3 PRECISION

The Captain — LambdaMART
(LightGBM) reranking

Directly optimizes NDCG using
lambda gradients (L08)



4 AGENTIC CRAG

Post-retrieval correction based on
user constraints.

Slider feedback
(Min Sharpe, Max Drawdown)

triggers THOUGHT → ACTION
loop (L11)

Deep Learning Models

CurveEncoder

1D-CNN



Processes equity curve time series into dense embeddings.

```
Input: equity curve (252 timesteps)
→ Conv1d(1→64, k=5) + BN + ReLU
→ Conv1d(64→128, k=5) + BN + ReLU
→ Global Avg Pool
→ Linear(128→128)
→ L2-normalize
→ 128-dim embedding
```

CURVE → VECTOR

TextBiEncoder

Scout, L06



Encodes strategy text descriptions into searchable vectors.

```
Base: all-MiniLM-L6-v2 (384-dim)
Frozen transformer +
trainable projection head
(Linear→ReLU→Linear→128-dim)
```

```
Trained with InfoNCE
contrastive loss (L06/L08)
```

```
Encodes query & doc independently
→ FAISS IndexFlatIP
```

TEXT → VECTOR

CrossEncoderJudge

Teacher, L07



Jointly scores (query, doc) pairs with metrics augmentation for distillation labels.

ms-marco-MiniLM-L-6-v2

```
Jointly processes
(query, doc) pairs with
metrics augmentation:
```

```
[CLS] query [SEP] text.
Metrics: Sharpe 1.5,
Drawdown -10% [SEP]
```

```
Used for Margin-MSE
distillation labels (L08)
```

PAIR → SCORE

Training Pipeline: InfoNCE + Margin-MSE Distillation

3

epochs

9,624

total steps

64 × 2

Batch × accum

2e-3

LR + warmup

🔧 InfoNCE Loss (L06/L08)

Contrastive loss for both [CurveEncoder](#) and [TextBiEncoder](#).

$$L = -\log(\exp(\text{sim}(q, k^+)/\tau) / \sum_j \exp(\text{sim}(q, k_j)/\tau))$$

Temperature $\tau = 0.07$

In-batch negatives: all other items in batch serve as negatives

Optimizes embedding space so similar strategies cluster together

🎓 Margin-MSE Distillation (L08)

Knowledge distillation from [Cross-encoder \(Teacher\)](#) to [Bi-encoder \(Student\)](#).

$$L = \text{MSE}((s_{\text{pos}} - s_{\text{neg}})_{\text{student}}, (t_{\text{pos}} - t_{\text{neg}})_{\text{teacher}})$$

Teacher scores (query, positive) and (query, negative) pairs

Student learns to reproduce teacher ranking margins

Enables **100x faster inference** than cross-encoder alone

⚡ **AMP** (Automatic Mixed Precision) on GPU for 1.5–2x speedup. Hard negatives via vectorized metric-based sampling ($O(n)$ vs $O(n^2)$ BM25).

Key Technologies & Justification

TECHNOLOGY	ROLE	JUSTIFICATION
BM25 rank_bm25	Sparse lexical retrieval	Classic inverted index with TF-IDF scoring. Handles exact keyword matches that dense models miss. Zero training cost.
SPLADE nreimers/splade-cocondenser-ense	Learned sparse retrieval (L05)	Learned sparse representations via $\log(1+\text{ReLU}(\text{logit}))$. Bridges lexical gap — expands query terms the model learned are relevant. Complementary to BM25.
Sentence-Transformers all-MiniLM-L6-v2	Dense text Bi-encoder (L06)	384-dim embeddings, fine-tuned on 1B+ sentence pairs. Captures semantic similarity beyond keyword match. Pre-trained — only projection head trained.
FAISS IndexFlatIP	ANN vector search (L02/L05)	Facebook's library for efficient similarity search. Inner product = cosine similarity on L2-normalized vectors. Sub-millisecond search on 205K vectors.
LightGBM LambdaMART	Learning-to-Rank reranker (L08)	Gradient-boosted trees directly optimizing NDCG via lambda gradients. Handles heterogeneous features. 100 trees, fast inference (<1ms).
Agentic CRAG L11	User feedback integration	Post-retrieval correction: THOUGHT → ACTION loop. Penalizes candidates violating user constraints (Sharpe, Drawdown sliders). No re-training needed.

i Each technology maps to a specific lecture reference (L02–L11) from the Deep Learning for Search course.

Experimental Pipeline & Evaluation

1 Data Collection

205,292 real text documents from HuggingFace datasets ([ag_news](#), [squad](#), [twitter-financial-news-sentiment](#), [yelp_review_full](#)) mapped to trading strategy domain with synthetic metrics and GBM-simulated equity curves.

2 Index Building

BM25 inverted index, FAISS dense indices (text + curve), SPLADE sparse matrix.

3 Training

3 epochs of InfoNCE + Margin-MSE on 205K docs. [CurveEncoder](#) (1D-CNN) and [TextBiEncoder](#) projection head trained.

4 Captain Training

LambdaMART on [300](#) pseudo-labeled queries × [50](#) candidates with 6 LTR features.

5 Comparative Evaluation

8 retrieval approaches benchmarked on [200](#) evaluation queries.

Evaluation Metrics

- **NDCG@5, NDCG@10** — ranking quality (L01)
- **MRR** — first relevant document position
- **Recall@10** — fraction of relevant docs found
- **Precision@5** — fraction of top-5 that are relevant
- **Avg / P95 / P99 Latency** — query timing

Validation Protocol

200 queries with ground-truth relevance via BM25 weak supervision + Sharpe-based nearest neighbors.

Comparative Study: 8 Approaches Benchmarking

APPROACH	NDCG@5	NDCG@10	MRR	RECALL@10	P@5	LATENCY
🏆 Iter2_RRF+Captain+Crag	0.612	0.674	0.713	0.682	0.574	28ms
Iter2_RRF+Captain	0.563	0.621	0.661	0.634	0.518	22ms
Iter2_RRF_All	0.501	0.558	0.594	0.571	0.462	18ms
Iter1_BM25+DenseText_RRF	0.458	0.517	0.549	0.533	0.421	14ms
Iter1_DenseText	0.398	0.461	0.492	0.471	0.363	8ms
Iter1_BM25	0.327	0.384	0.413	0.392	0.291	5ms
Iter1_SPLADE	0.354	0.412	0.438	0.418	0.313	10ms
Iter1_DenseCurve	0.241	0.293	0.327	0.269	0.198	12ms

💡 Key Finding

The full Alliance (RRF + Captain + CRAG) consistently outperforms all individual systems. RRF fusion alone improves NDCG@10 by **15-20%** over best single system. The Captain (LambdaMART) adds another **8-12%** by learning optimal feature weights. Agentic CRAG provides additional precision for constrained queries.

Hardware Requirements

DEVELOPMENT

GPU	NVIDIA CUDA 12.1+ (RTX 3060/4060, 6-8 GB VRAM)
RAM	16-32 GB
Storage	~2 GB (parquet + indices + embeddings)
Training Time	~30-60 min for 3 epochs on GPU
Bottleneck	Cross-encoder teacher inference during Margin-MSE (GPU-only)

INFERENCE

CPU (ACCEPTABLE)	
CPU / RAM	8+ cores / 4-8 GB
Latency	<200ms full pipeline on CPU
GPU (OPTIONAL)	
Speedup	Text encoding: ~50ms → ~5ms per query
FAISS	Sub-millisecond on CPU for 205K vectors

COMPONENT COMPARISON

COMPONENT	DEVELOPMENT (GPU)	INFERENCE (CPU)
BM25 search	Fast	Fast
Dense encoding	GPU required	~50ms / batch
FAISS search	Sub-ms	Sub-ms
LambdaMART	<1ms	<1ms
CRAG filter	<1ms	<1ms

Thank You

Questions?

- 01 Multimodal fusion (sparse + dense + curve) significantly outperforms any single retrieval approach.
- 02 LambdaMART reranking and Agentic CRAG add meaningful precision improvements.
- 03 The system achieves <200ms query latency on CPU at inference time — suitable for production deployment.